

EN3160 Fundamentals of Image Processing and Computer Vision

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Electronic and Telecommunication Engineering

Course Outline

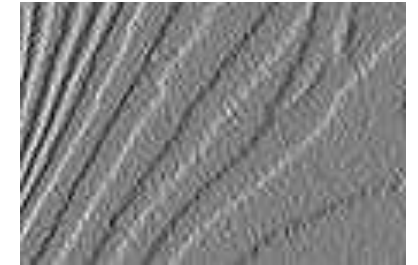
1. Image processing and early vision
 - Sampling, interpolation, point operations, filtering (e.g., convolutions), edge detection, feature extraction, optical flow
2. Fitting and alignment
 - Least squares, voting methods, transforms, stitching
3. Image formation
 - Perspective projection, cameras, light and shading, color
4. 3D vision
 - Camera calibration, two-view geometry, structure from motion, stereo, dense 3D reconstruction (neural radiance fields, Gaussian splatting)

5. Segmentation
 - Thresholding, region growing, snakes, grab cuts, introduction to deep learning based methods
6. Introduction to deep learning for vision
 - Linear classifiers, image classification, object detection, generative methods (auto-regressive, diffusion)
7. Recent topics (time permitting)

Image Processing and Early Vision

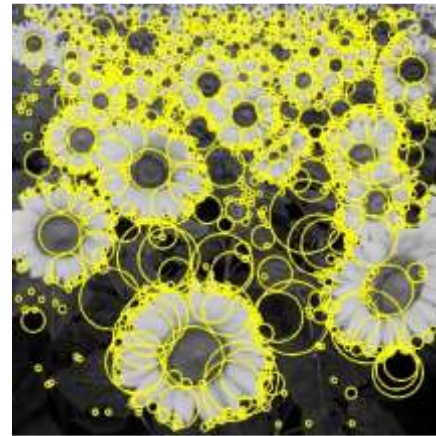


$$\star \text{ [Gaussian Kernel] } =$$

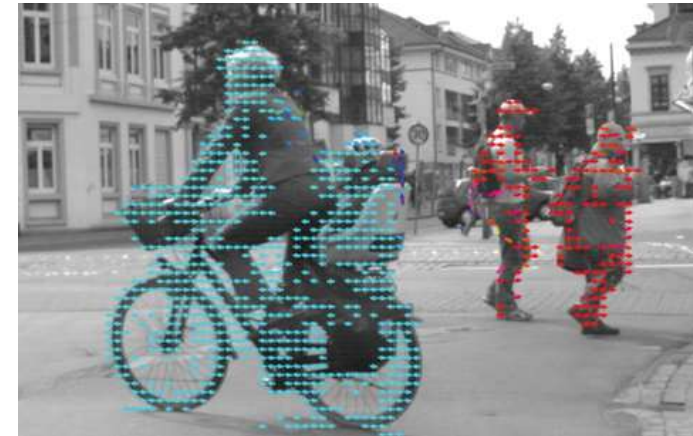


Sampling, interpolation, Fourier analysis

Linear filtering, edge detection

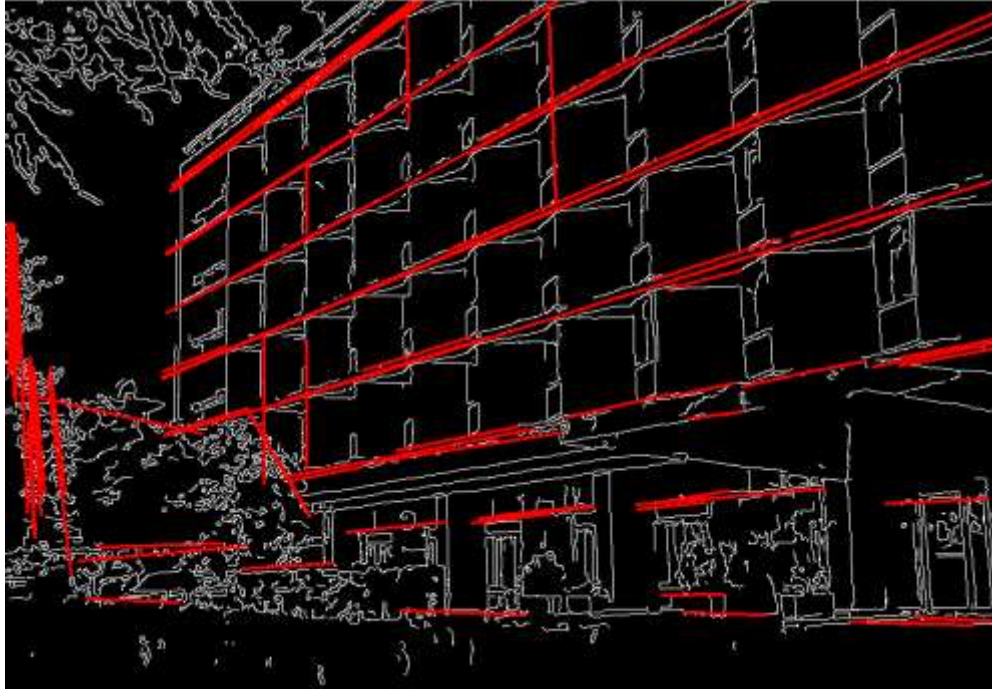


Feature extraction



Optical flow

Fitting and Alignment

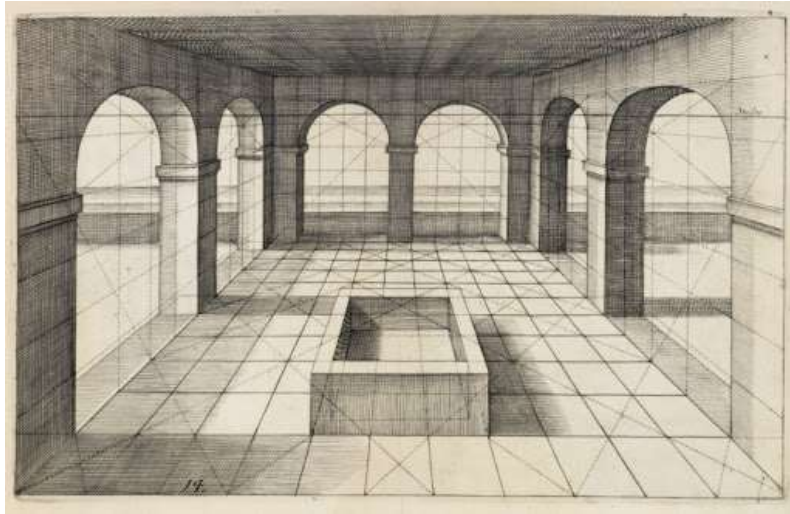


Fitting: Least squares, voting methods



Alignment, image stitching

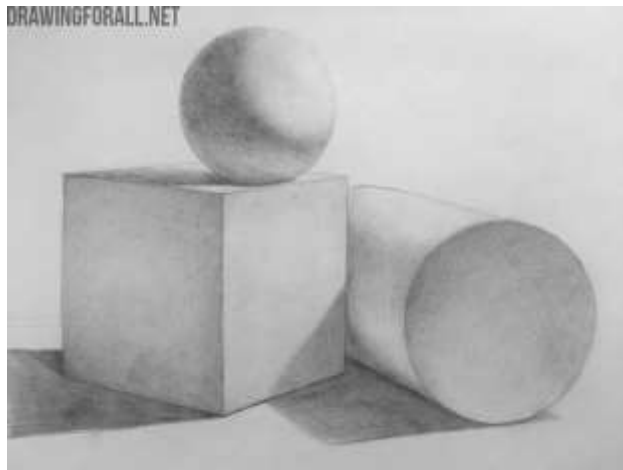
Image Formation



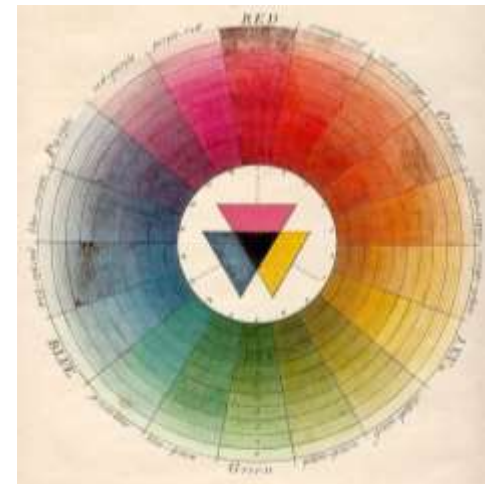
Perspective projection



Camera optics



Light and shading



Color

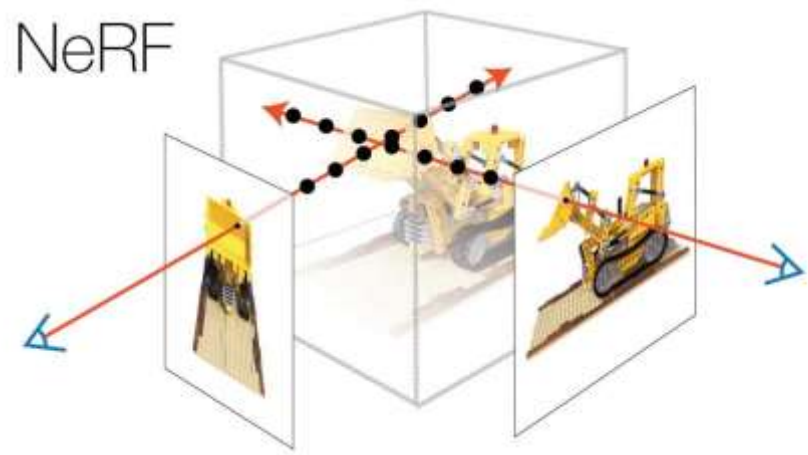
3D vision



Camera calibration



Two-view geometry, structure from motion Stereo



Images



Neural
Network
k

Reconstruction

Cameras, Depths, Points, and Correspondences



Light fields and dense 3D reconstruction

Wang et al, VGGT: Visual Geometry Grounded Transformer, CVPR 2025.

Segmentation

Segmentation Output (Semantic)

Input



Segmentation

Labeled



Binary Segmentation Output

Input



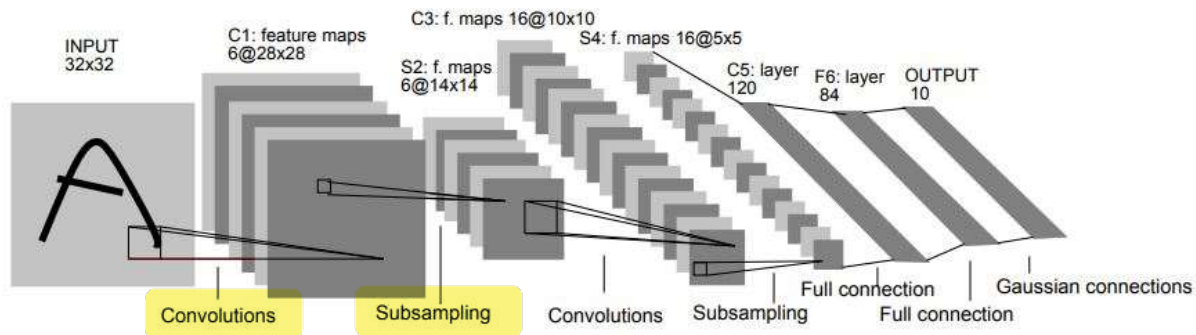
Segmentation

Binary image

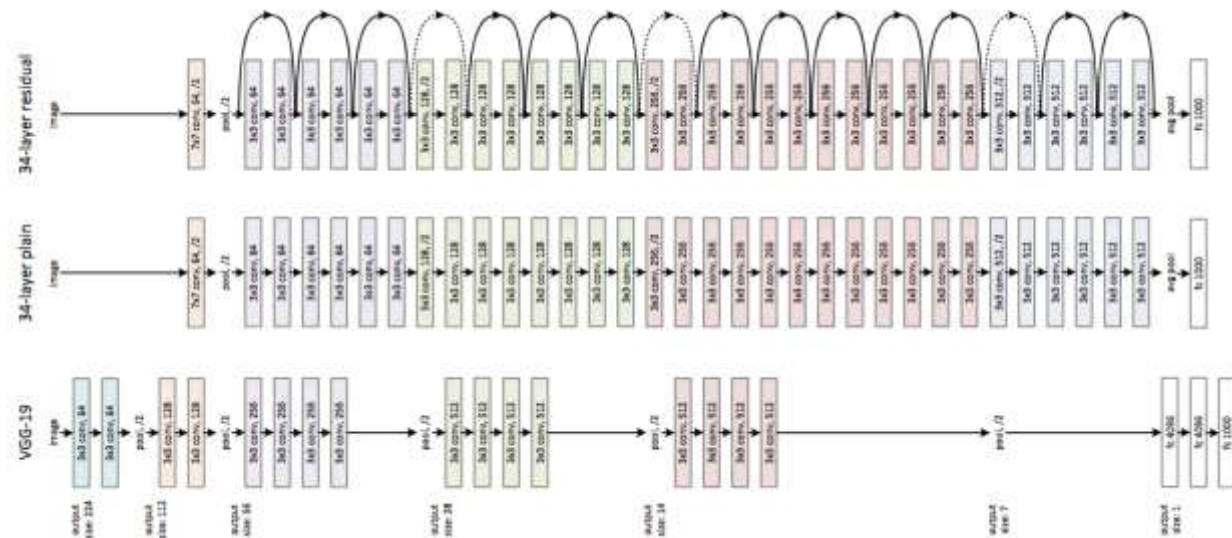


Ravi et al, SAM 2: Segment Anything in Images and Videos, ICLR 2025.

Deep Learning for Vision

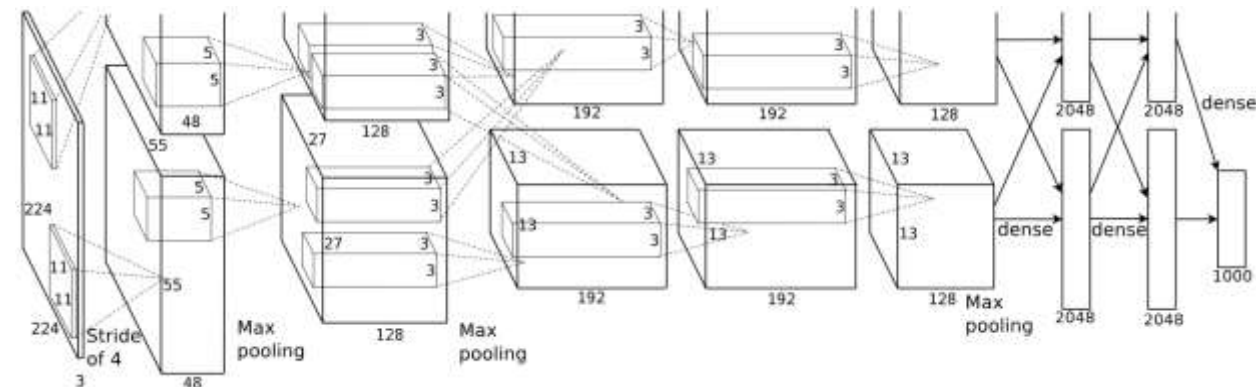


LeCun et al., Gradient-Based Learning Applied to Document Recognition, Proceedings of IEEE 1998.

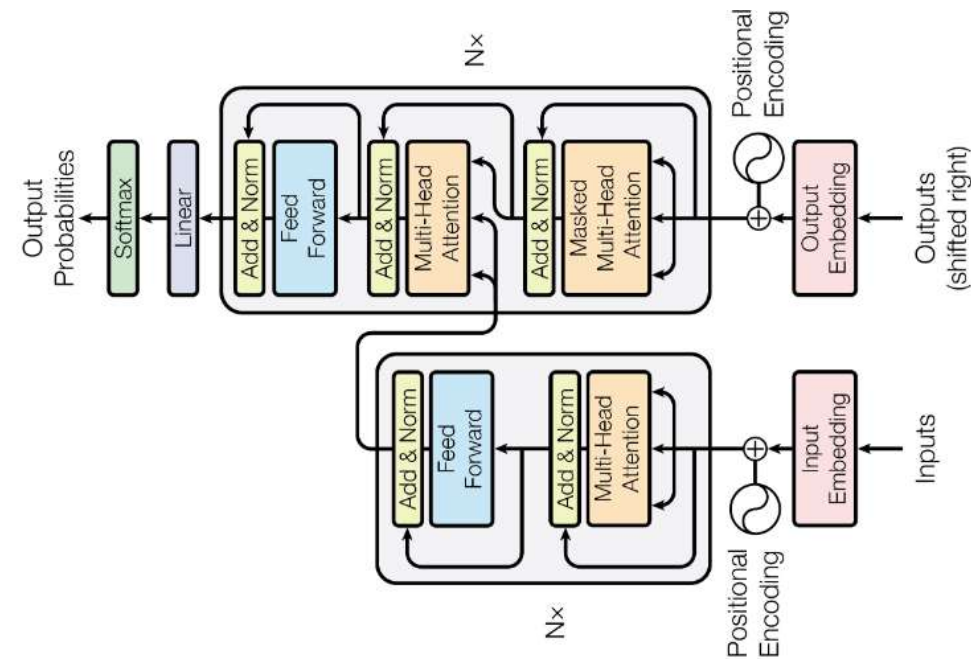


He et al., Deep Residual Learning for Image Recognition, CVPR 2016.

AlexNet

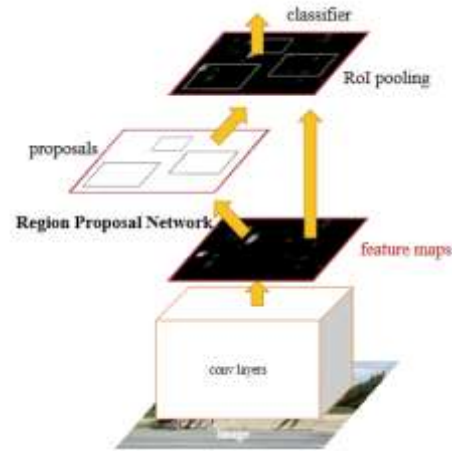


Krizhevsky et al., ImageNet Classification with Deep Convolutional Neural Networks, NeurIPS 2012.

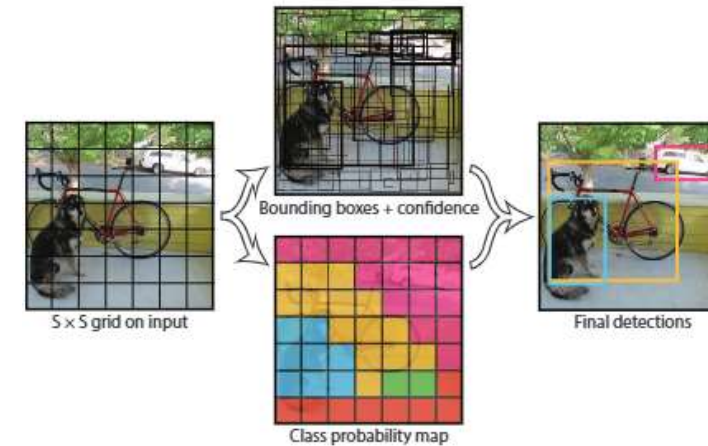


Vaswani et al., Attention Is All You Need, NeurIPS 2017.

Deep Learning for Vision

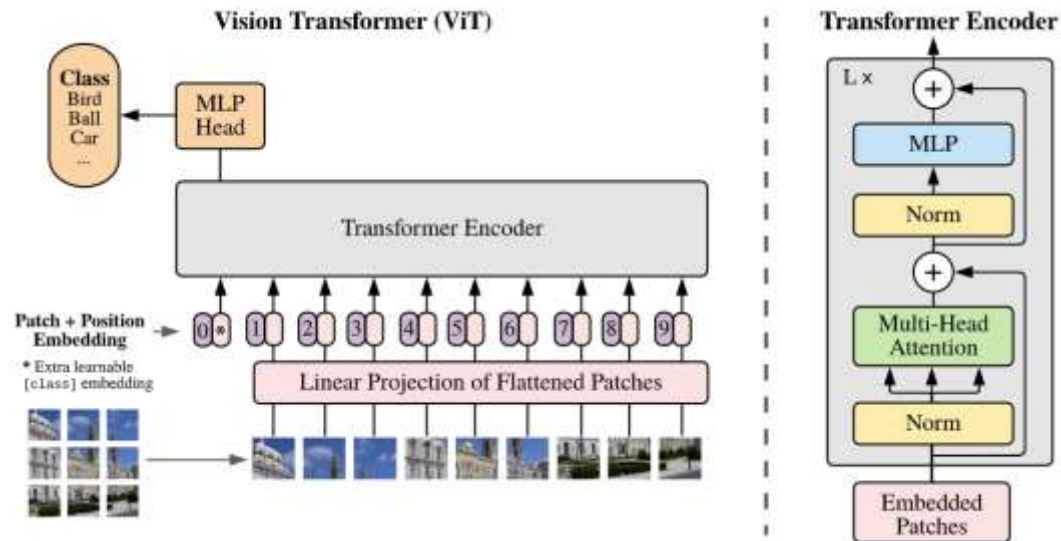


Ren et al, Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks, NeurIPS 2015.

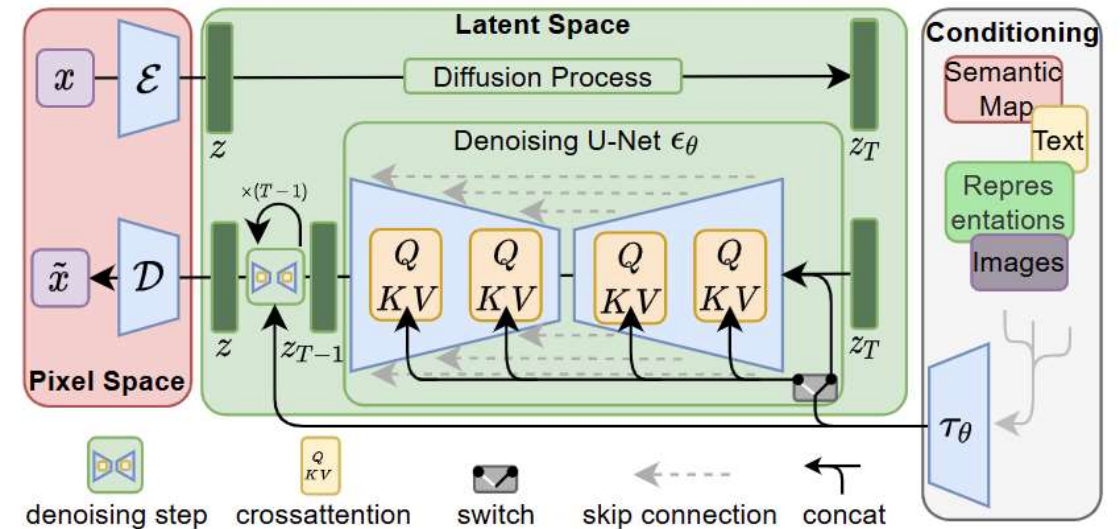


YOLO

Redmon et al., You Only Look Once: Unified, Real-Time Object Detection, CVPY 2016.



Dosovitskiy et al., An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale, ICLR 2021.



Rombach et al., High-Resolution Image Synthesis with Latent Diffusion Models, CVPR 2022.

Agenda

- | | | |
|---|--|---|
| 1 | Image processing and computer vision | |
| 2 | Introduction to AI, ML, and Deep Learning, AI, | What is machine learning, neural networks, and deep learning?
Convolutional Neural Networks (CNNs)
Attention and transformers |
| 3 | Introductions to Generative AI | Introduction to generative AI
Introduction to self-supervision
Generative AI applications |
| 4 | ENTC Vision Projects | Projects done at the department |

What Is Image Processing?

- In digital image processing, we manipulate a digital image to produce another digital image, which is an enhanced version of the original image.

- E.g., blurring, noise filter



Input and Output both are images



(a) Noisy image



(b) Gaussian filtered image



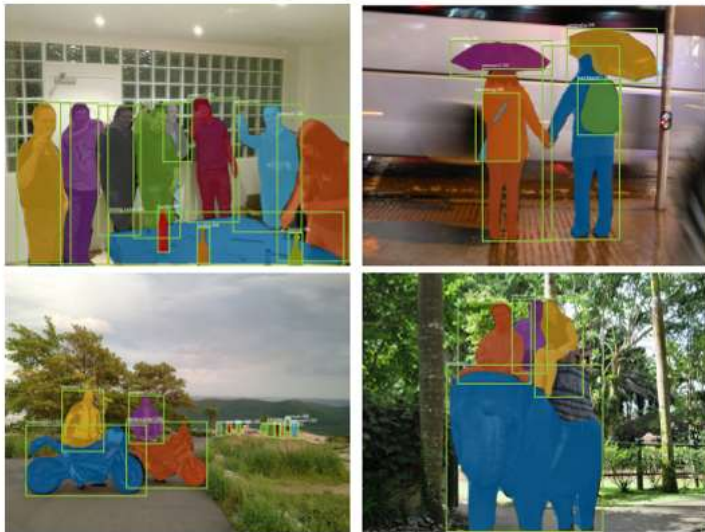
(c) Image with salt and pepper noise



(d) Median filtered image

What is Computer Vision?

- In computer vision, we analyze a digital images or videos to make a decision.
- E.g., face detection, object detection, semantic segmentation.



Instance segmentation in Mask R-CNN

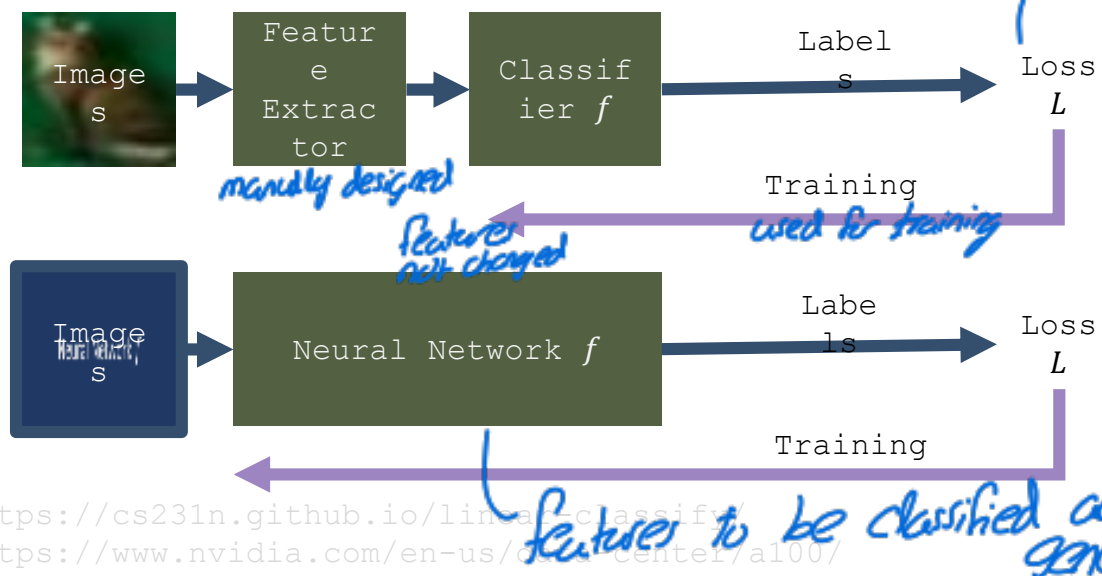
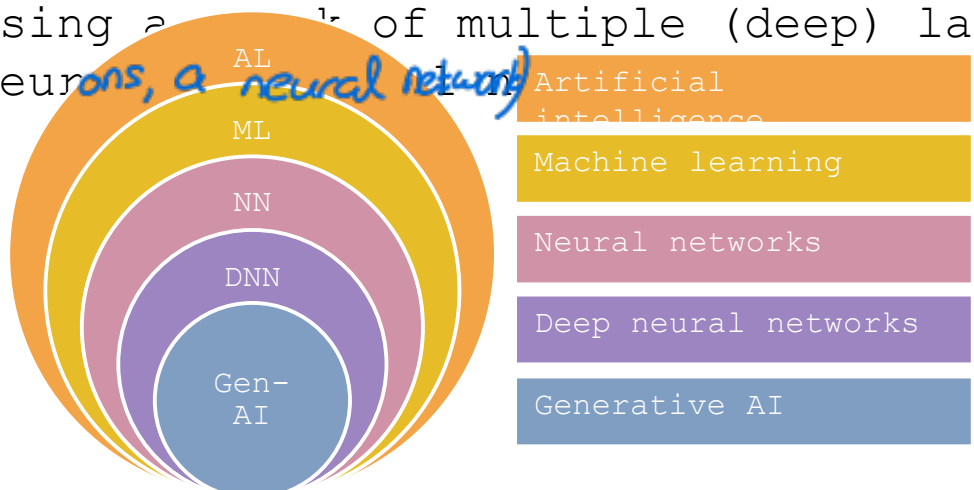


Face detection (Wikipedia)

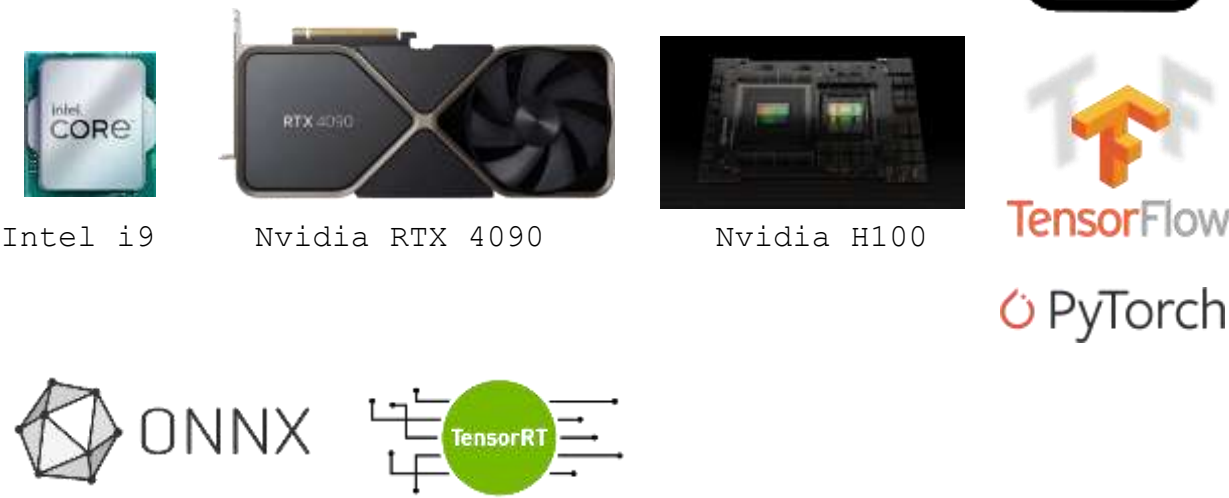


Google Whisk: Generate an image of a beautiful and serene scene of a young man asking for the hand of a beautiful girl!¹³

Deep learning is a machine learning advancement that allows the system to automatically discover (learn) **representations** (representation learning) using a stack of multiple (deep) layers (of neurons, a neural network)

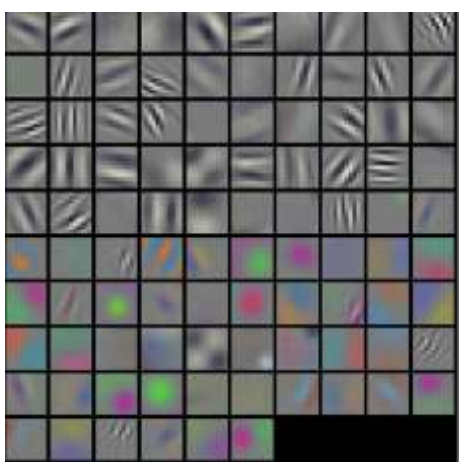


Research → industry
Computation power



DISRUPTIVE ADVANCEMENTS

- Perceptron 1958
- Convolutional neural networks 2012
- GPU based parallelization
- Transformers 2017
- Generative pre-trained models 2022



AlexNet layer 1, 11 × 11 weights

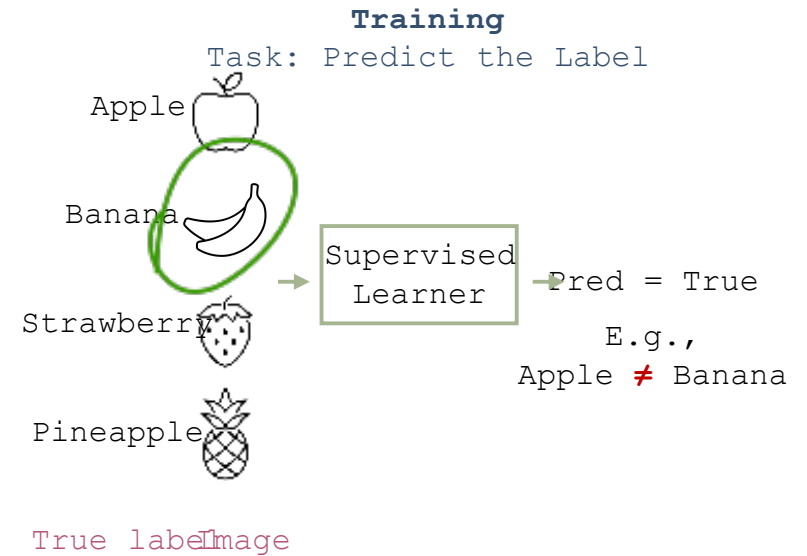
Major Areas of Application of AI

Area	Examples	Key Technologies	Example Use Cases
Computer Vision	Tesla's self-driving Avatar 2 production	CNNs, Transformers	Defect detection Volume application VR displaying Self driving
Natural Language Processing (NLP)	Google search, Google's Bard OpenAI's ChatGPT Meta's Llama2	Transformers	Public chatbots Negotiators
Data Analytics	Amazon's product recommender system YouTube's video recommended system Quantitative finance models of a hedge fund	Machine learning, big data, NLP, cloud, visualizations	Recommender systems Customer profiling Business analytics



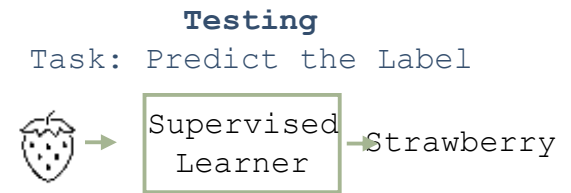
Supervised

Learns from labeled examples, like a student studying with answers provided. Needs pre-labeled data (think flashcards).



Self-supervised

Trains on unlabeled data by creating its own supervisory signals. Discovers patterns by itself. Two common approaches: reconstruction, contrastive learning.



Reinforcement

Learns through trial and error, like an animal receiving rewards for desired actions. Interacts with an environment and gets

feedback (rewards/penalties) to learn

There is a reward function

Supervised

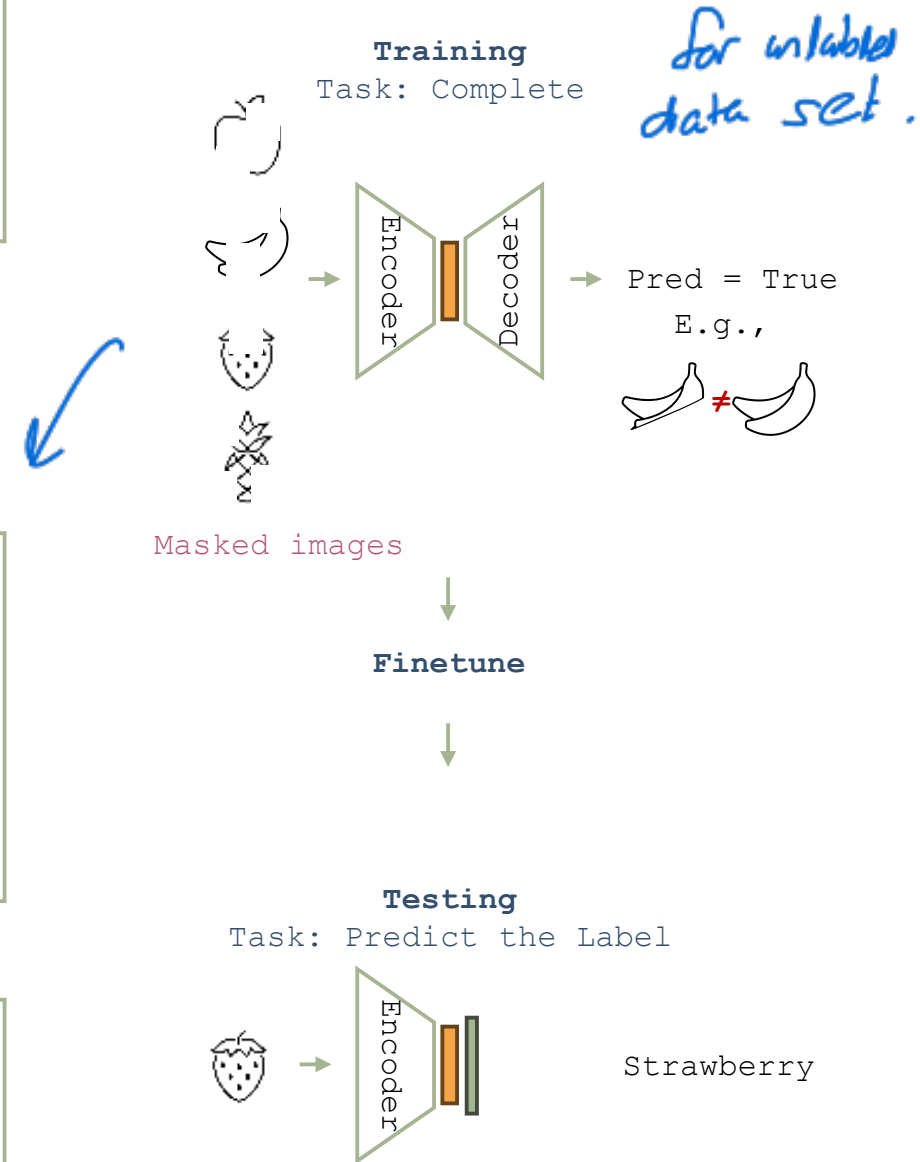
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Self-supervised

Trains on unlabeled data by creating its own supervisory signals. Discovers patterns by itself. Two common approaches:

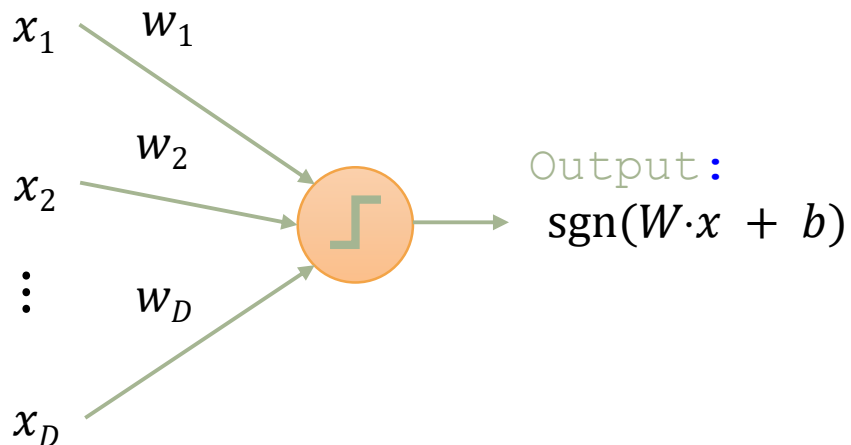
reconstruction,
contrastive learning.

Reinforcement

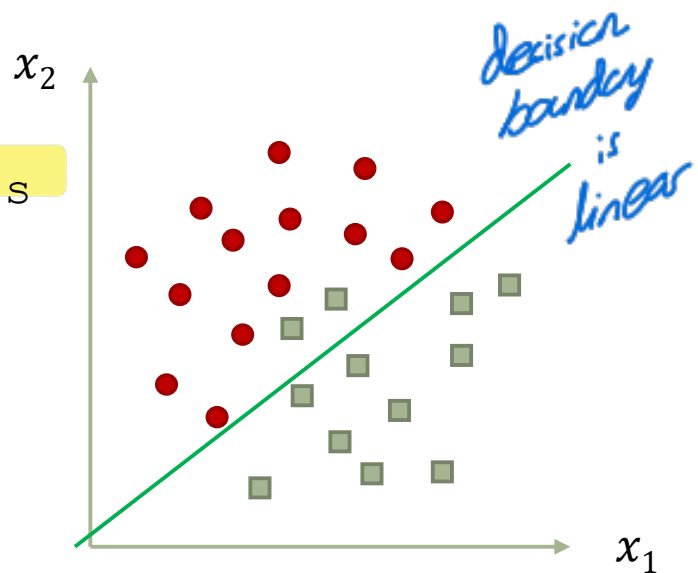


The Perceptron

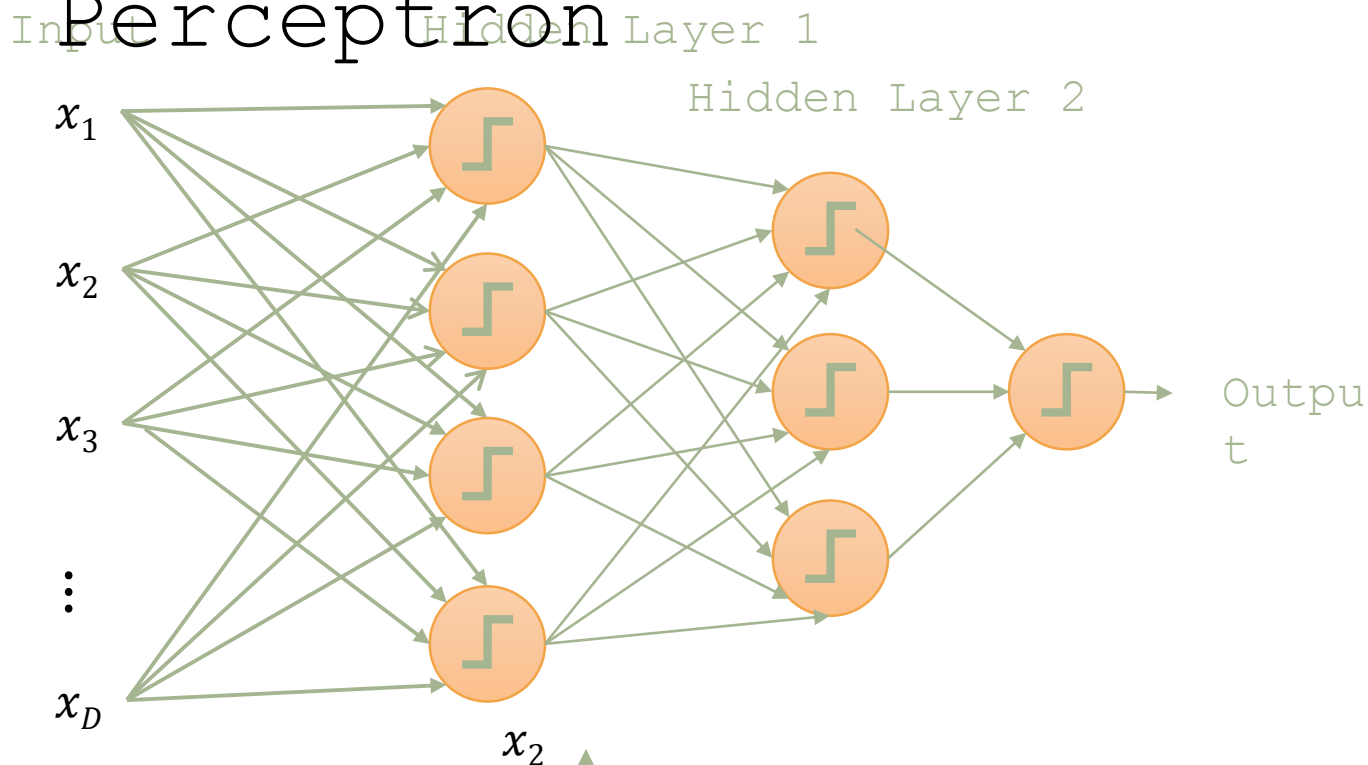
Input Weights



Perceptron is
a linear
classifier.

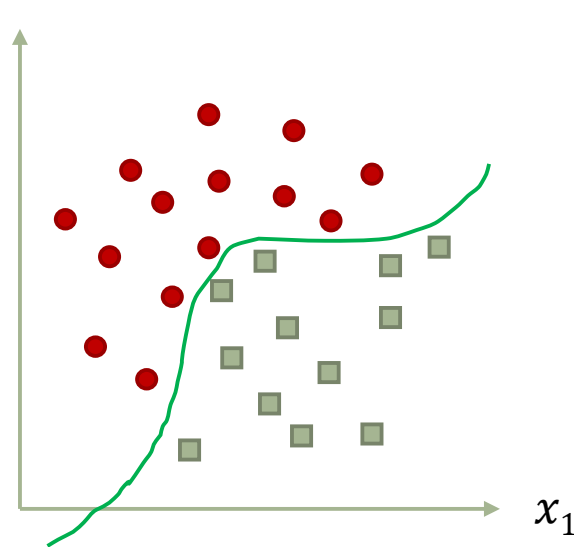


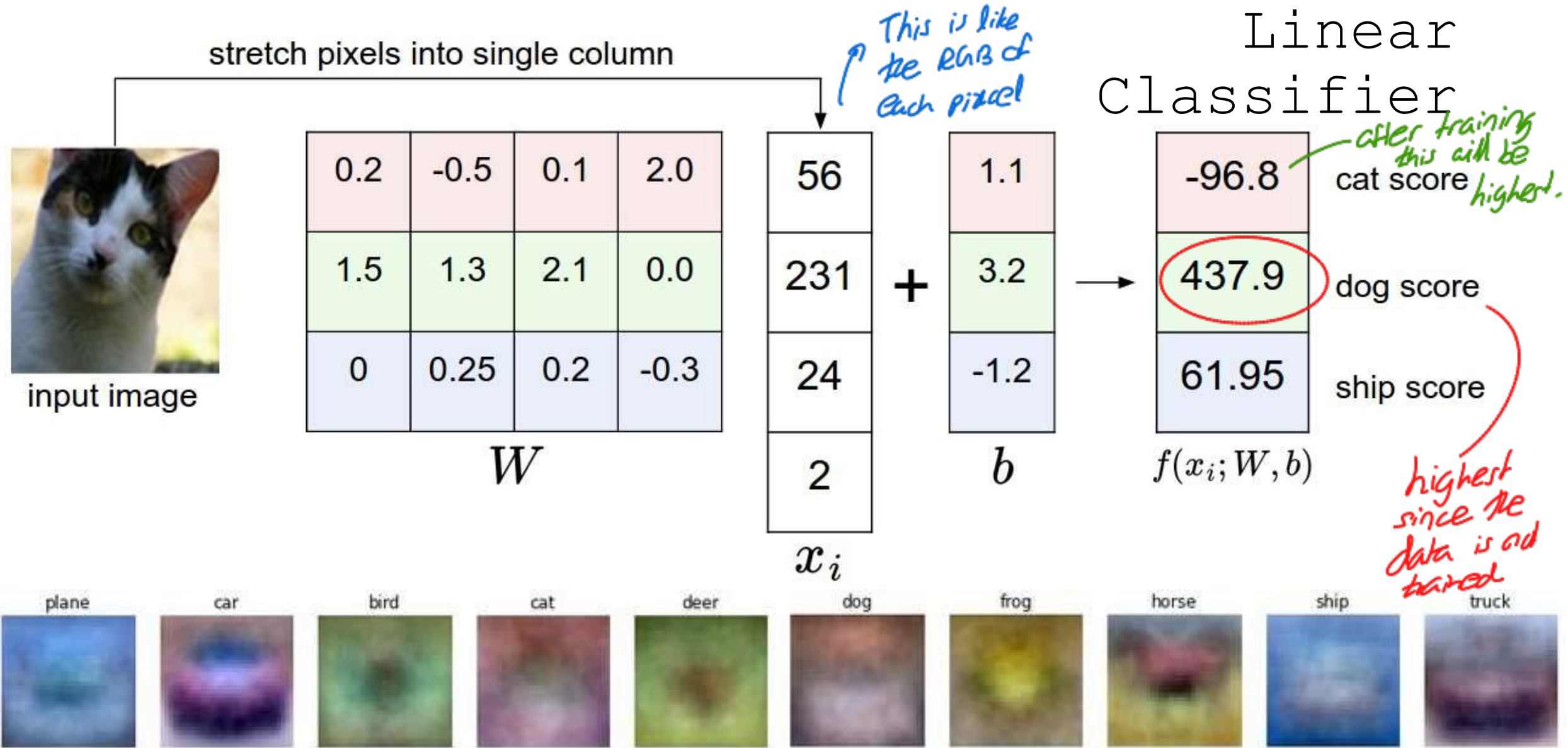
A Multi-Layer Perceptron



Can learn
nonlinear
functions

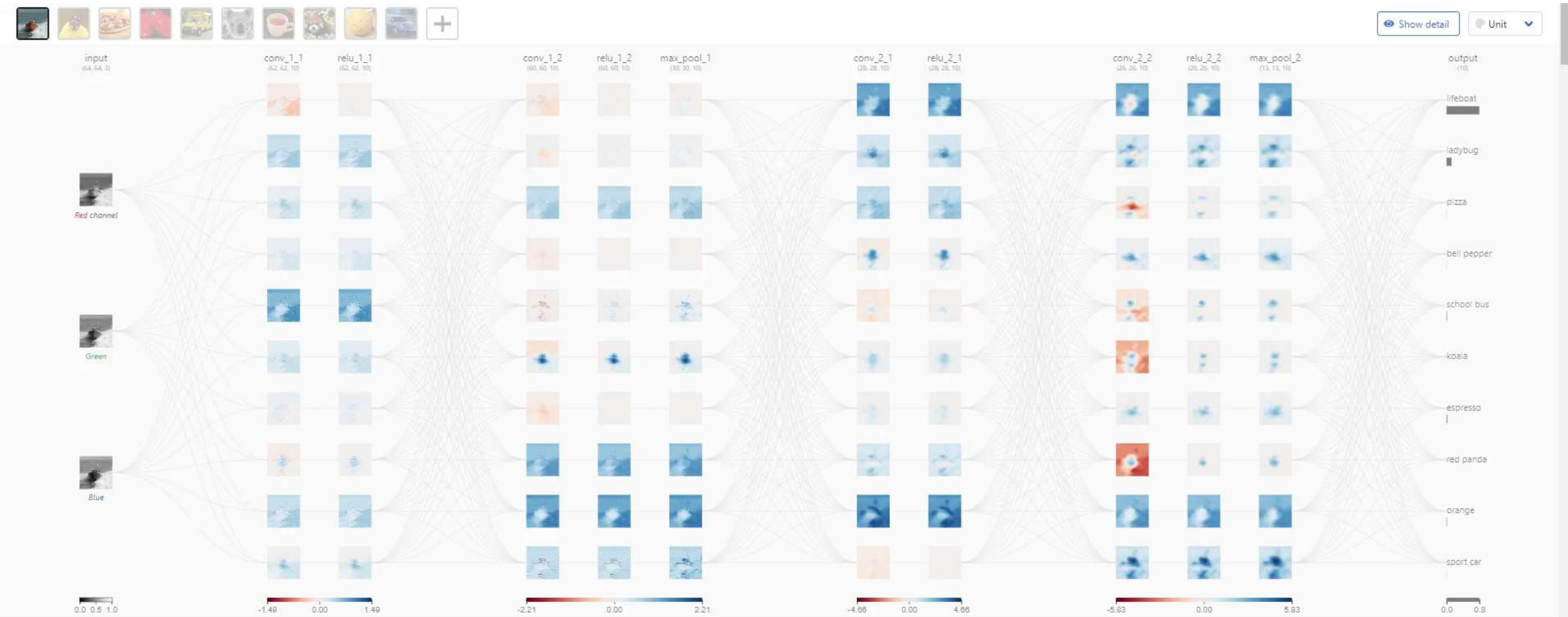
provided each
perceptron has
a
differentiable
nonlinearity.



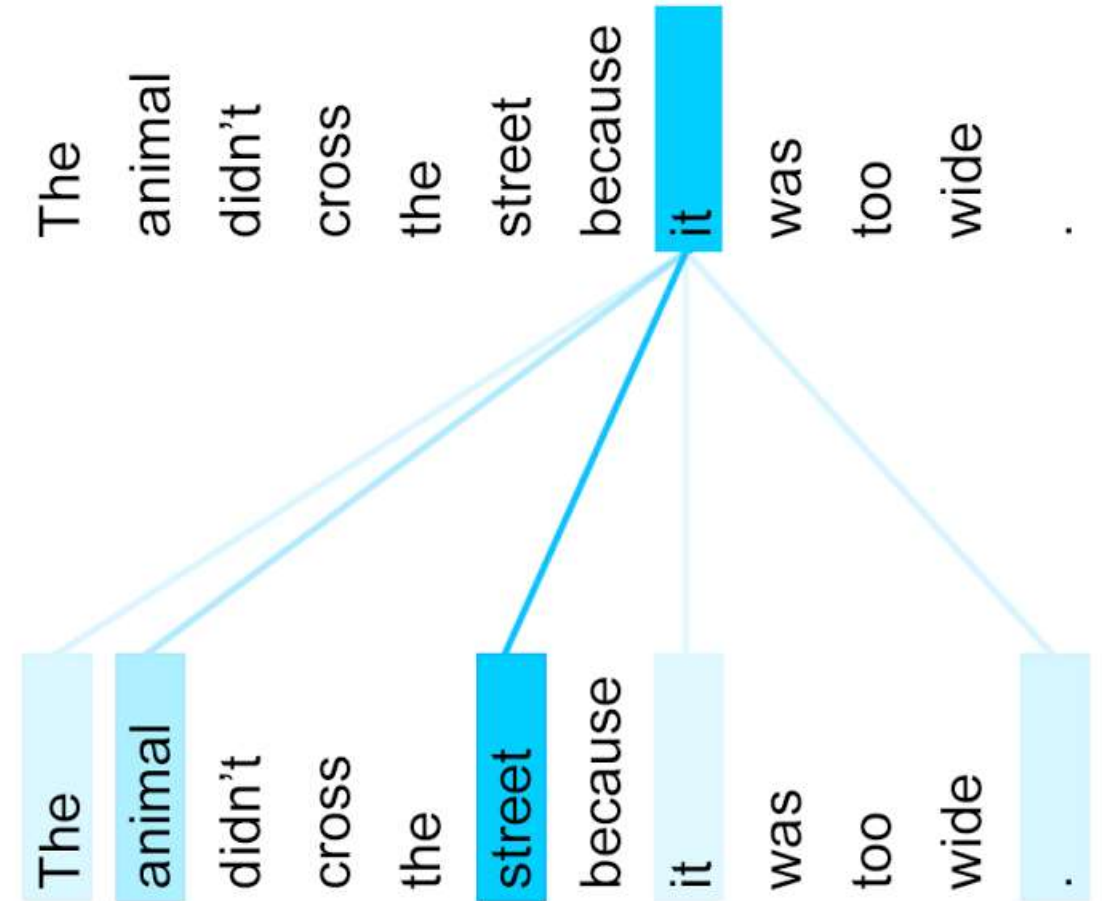
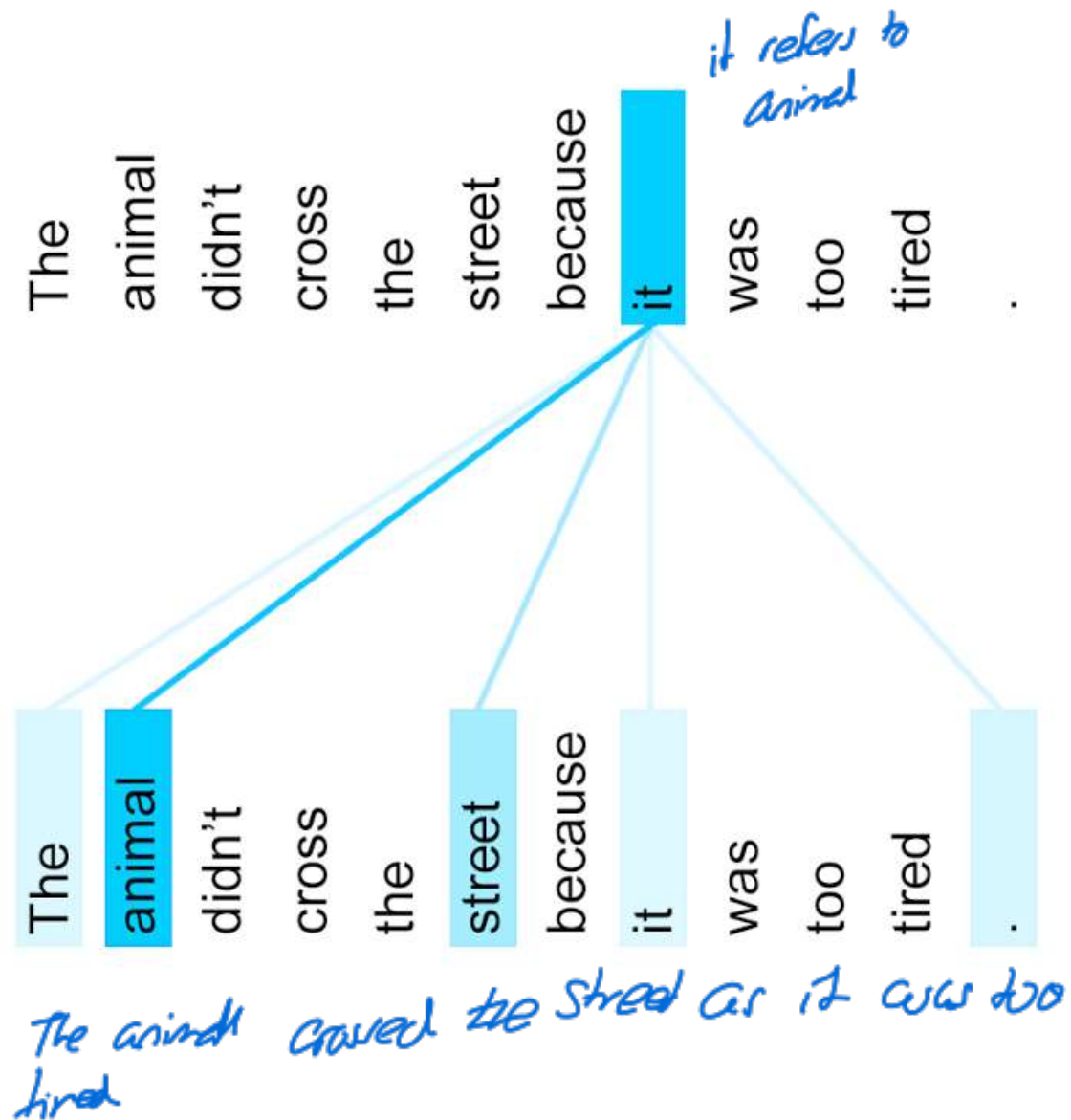


Example learned weights at the end of learning for CIFAR-10. Note that, for example, the ship template contains a lot of blue pixels as expected. This template will therefore give a high score once it is matched against images of ships on the ocean with an inner product.

CNNs in Image Classification

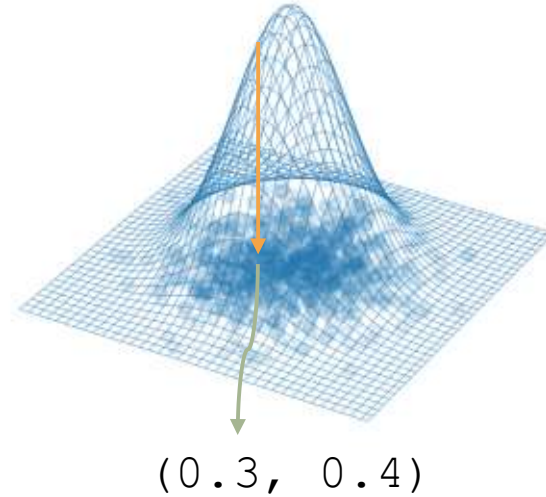
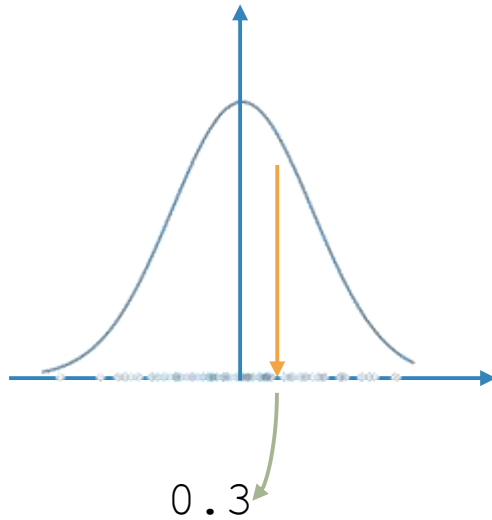


Attention (in Transformers) in NLP



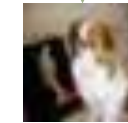
Generative Models

for an image this will be 30,000,000 images. when we get a point in the image it will look like an image

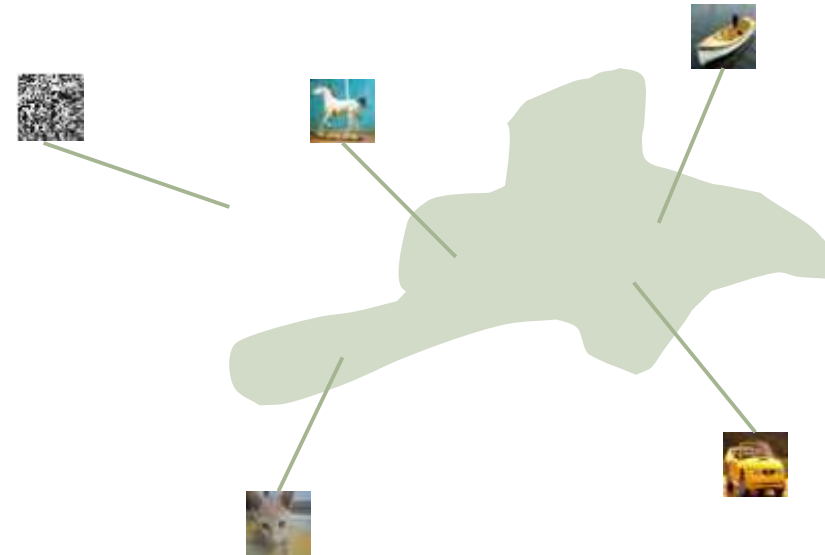


pdf representing
32x32 images

sample a point



- Generative models learn the underlying distribution of data.
- Once learned, they can generate new data samples that resemble the training data.



Generative AI

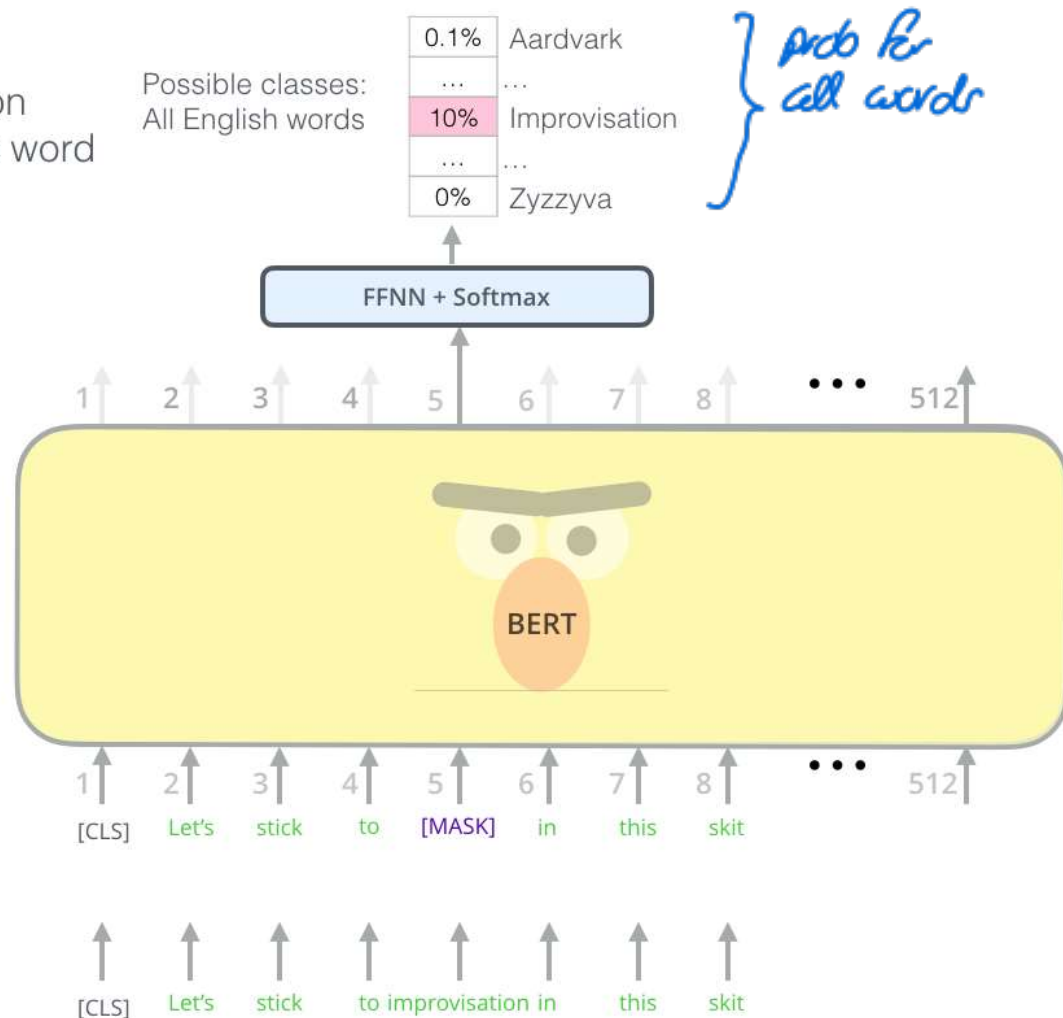
Generates new content, data, or information that is similar to, or mimics, human-created content.

Text Generation	Generative AI models like Generative Pre-trained Transformers (GPT) can generate human-like text	Chatbots, and content creation. E.g., ChatGPT, Gemini.
Image and Video Generation	Stable diffusion generates realistic images, art, and visual content based on a textual description	Advertising and entertainment industry. E.g., MidJourney
Data Augmentation	Generative AI can generate additional data samples to augment datasets for training machine learning models	Training models with less actual data
Music Composition	AI systems can generate music compositions in various styles and genres, based on patterns learned from existing music.	Entertainment industry

BERT

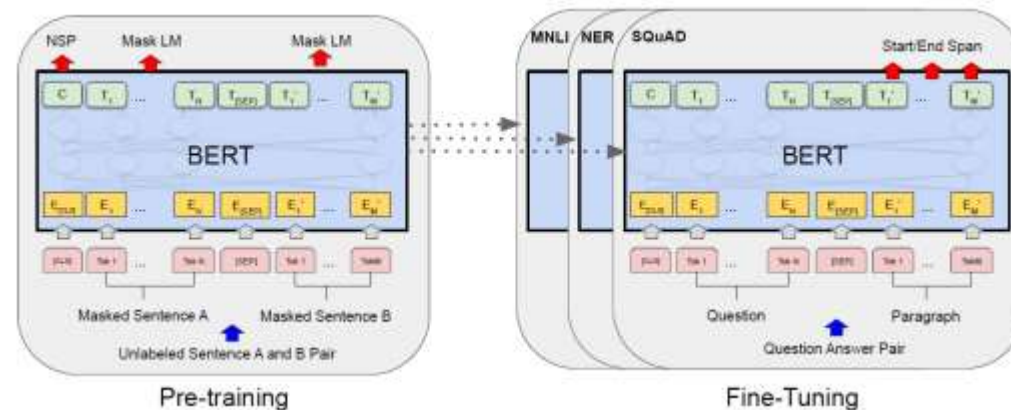
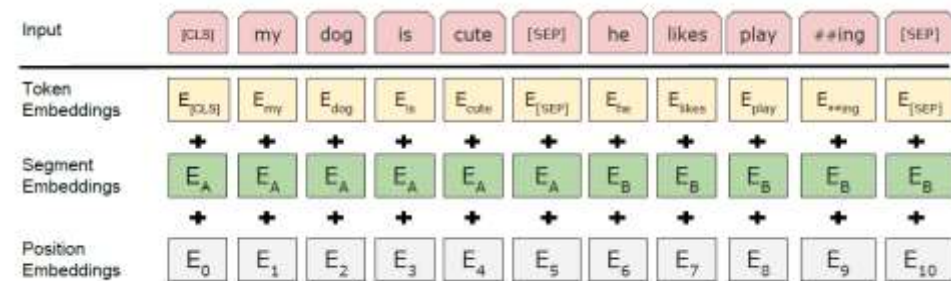
Devlin et al., BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL 2019.

Use the output of the masked word's position to predict the masked word



Randomly mask 15% of tokens

Input

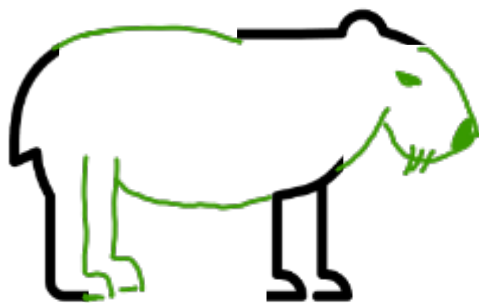


Fun Exercise: What is Hidden?

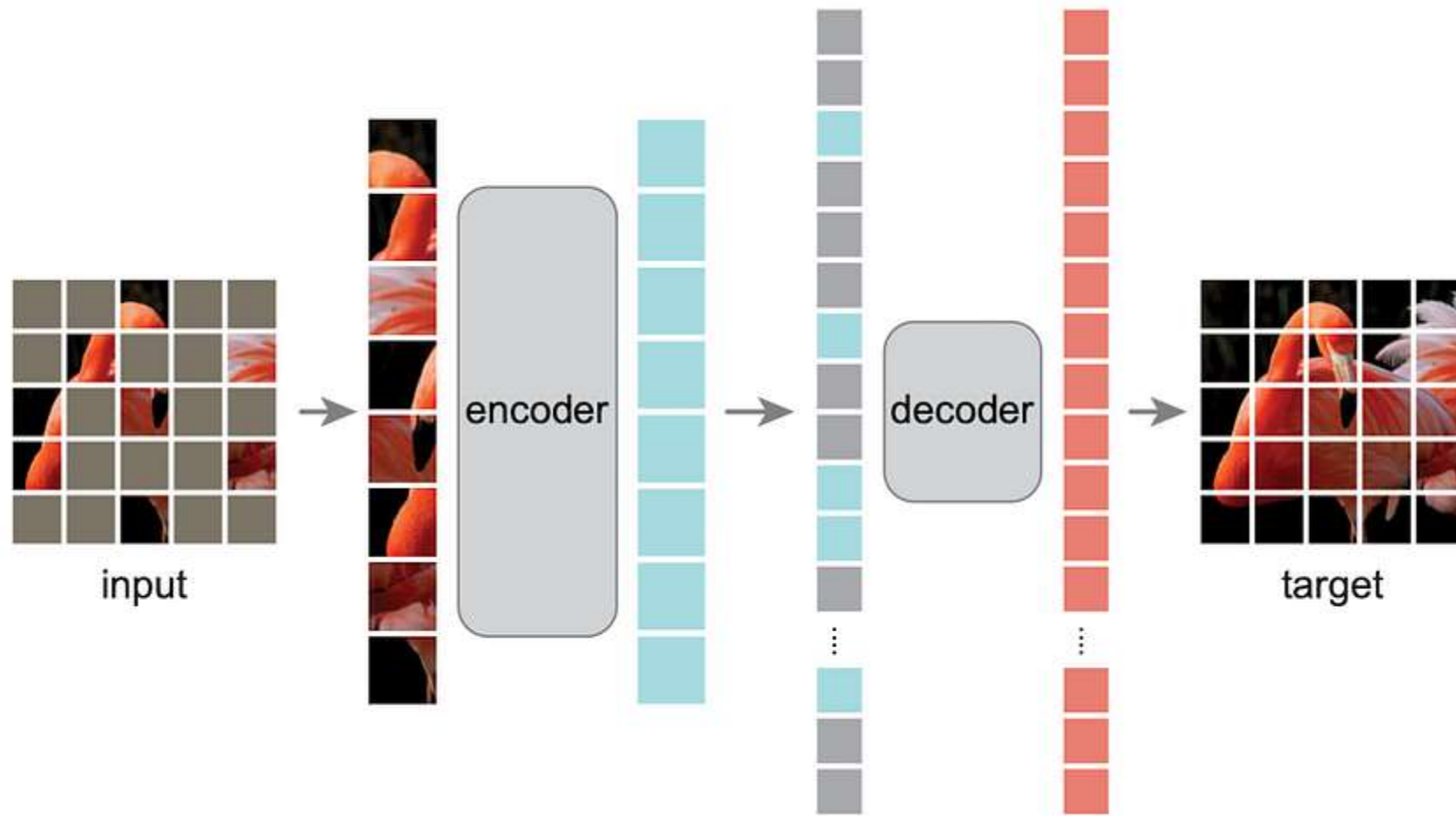


*we have the actual
image.*

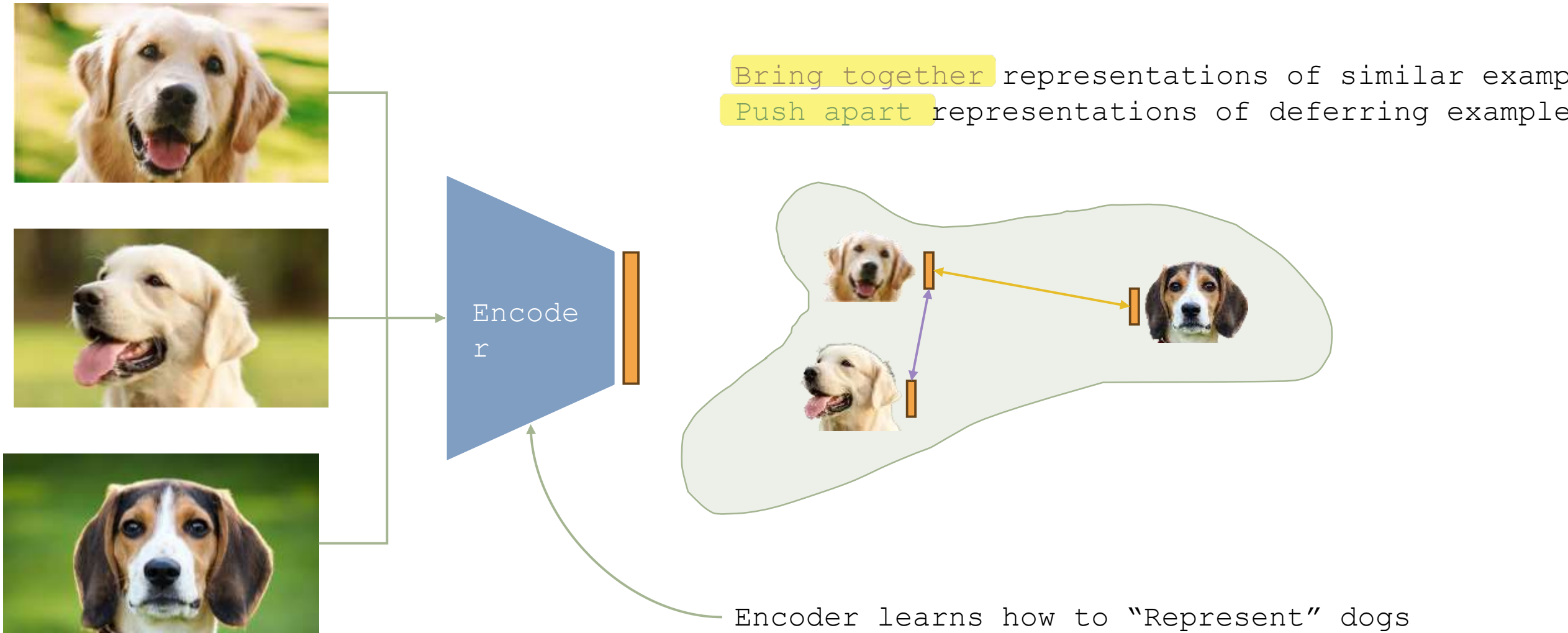
*Then we mask it.
Then ask the model to
predict.*



Self-Supervised Pre-Training: Masked Auto Encoder

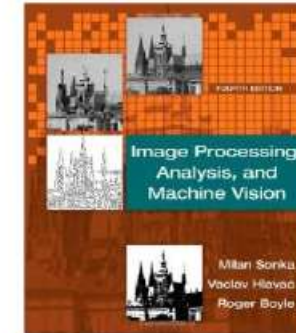
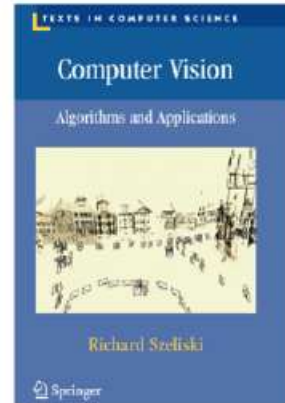
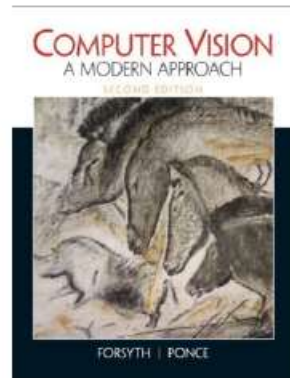
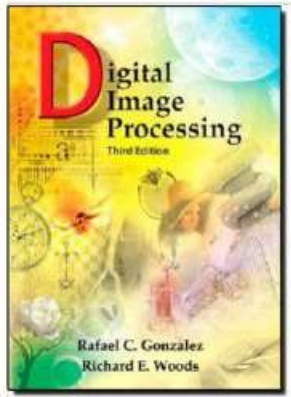


Contrastive Learning



Text Books, Journals, and Conferences

1. Gonzalez and Woods, Digital Image Processing
2. Forsyth and Ponce, Computer Vision: A Modern Approach
3. Richard Szeliski, Computer Vision: Algorithms and Applications (available online)
4. Milan Sonka, Image Processing, Analysis, and Machine Vision



Conferences: CVPR, ICCV, ECCV, WACV, BMVC, ICIP

Journals: PAMI, IJCV, Pattern Recognition, TIP, Cybernetics, Computer Vision and Image Understanding

Tips for Successful Completion

1. Mindset: "I want to solve this vision problem. What are the tools available? How have others solved it? How should I go about solving it?"
2. Attend every single lecture.
3. Go through the material the night before and engage in a good discussion in class.
4. Implement at least one algorithm discussed in class before the next class. This is in addition to the assignments.

Software and Cameras for Industrial Vision

1. In EN3160:

- We use Python, OpenCV and PyTorch.

2. Computer vision software companies

- <https://www.cognex.com/>
- <https://www.mvtec.com/>

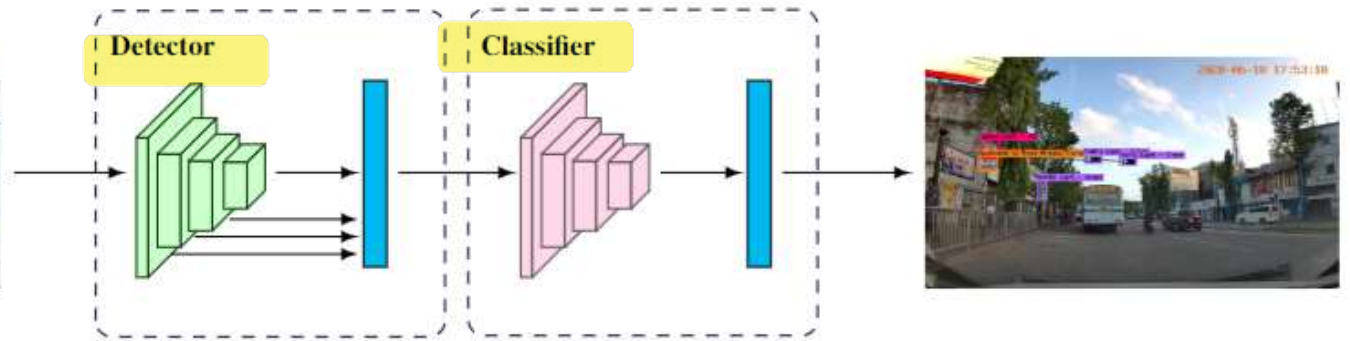
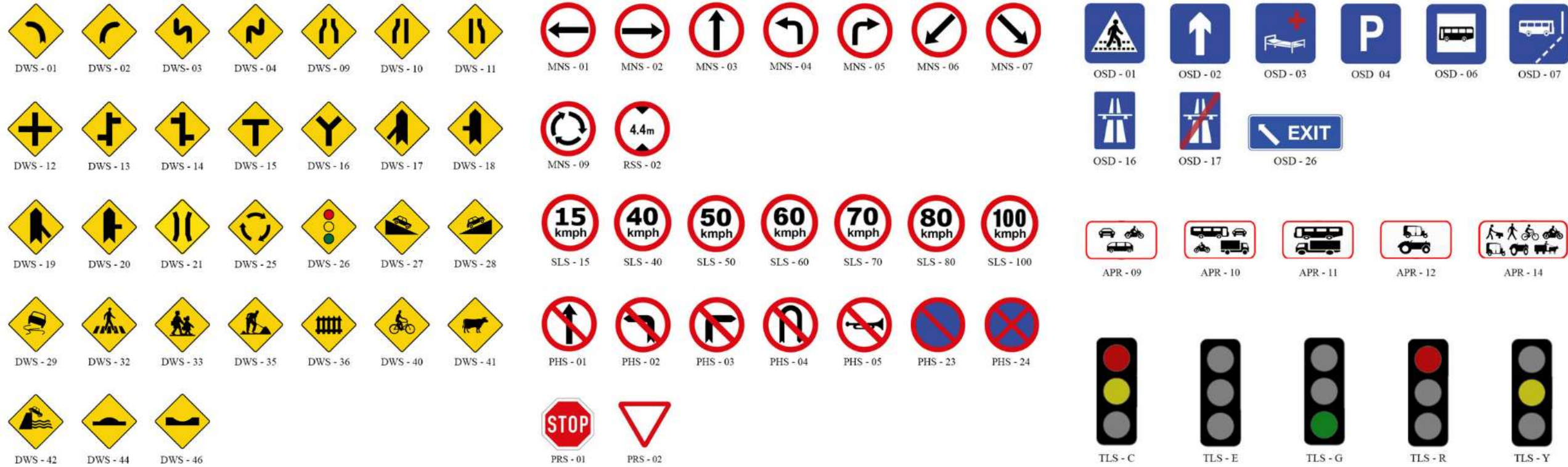
3. Camera companies

- <https://www.baslerweb.com/en/>
- <https://www.flir.com/iis/machine-vision/>

4. Camera catalogue

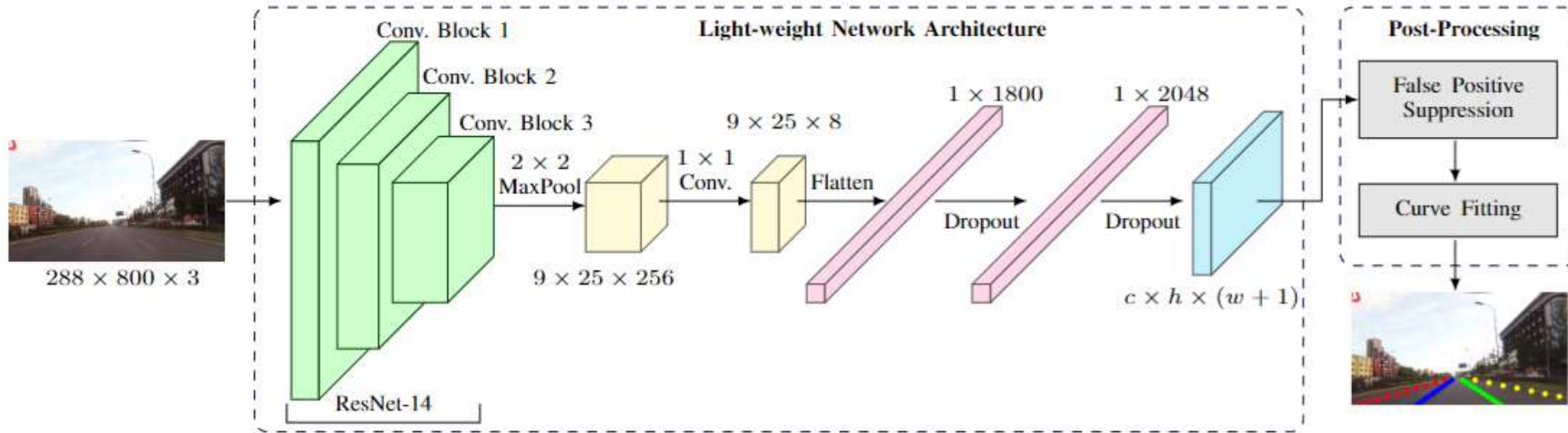
- <https://www.edmundoptics.com/c/cameras/1012/>

Traffic Sign and Traffic Light Detection on Embedded Systems

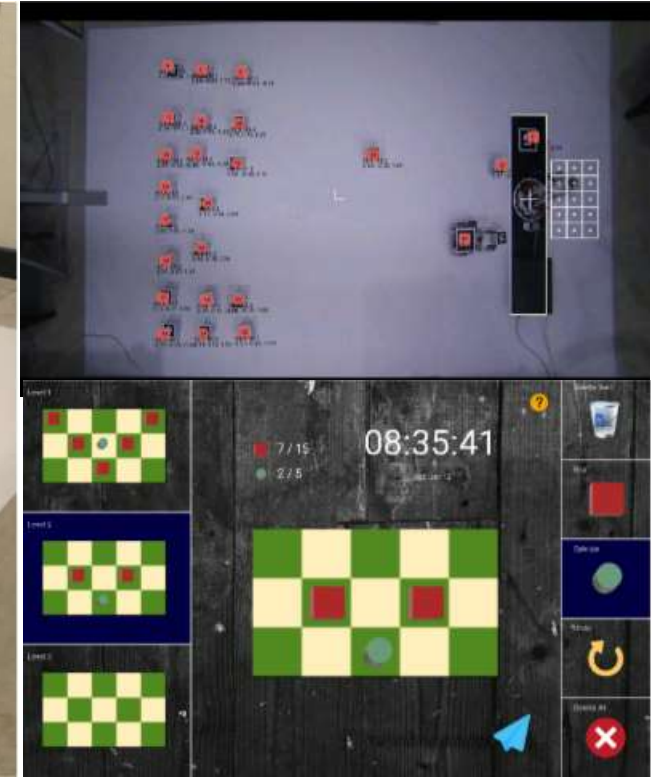
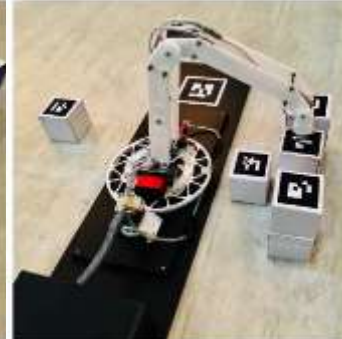
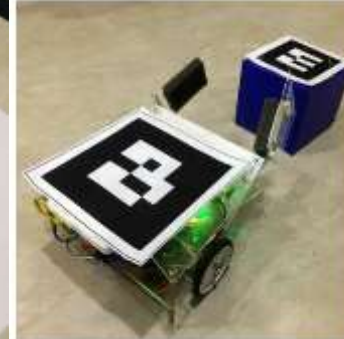
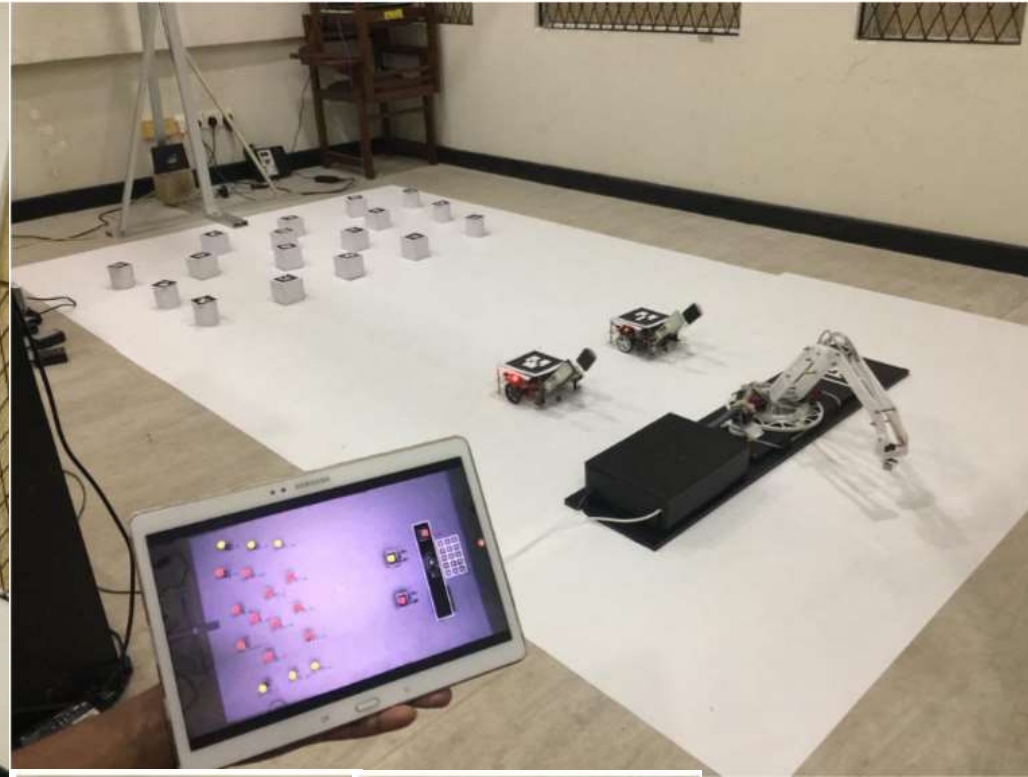
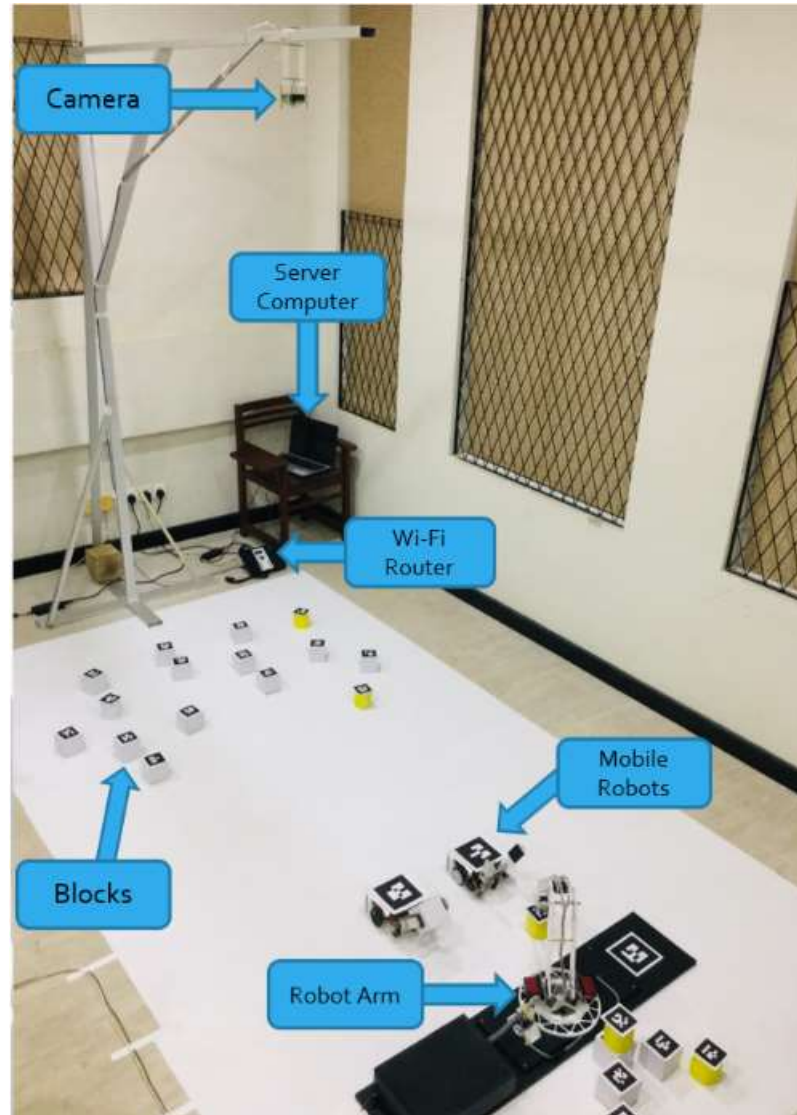


SwiftLane: Towards Fast and Efficient Lane

Detection

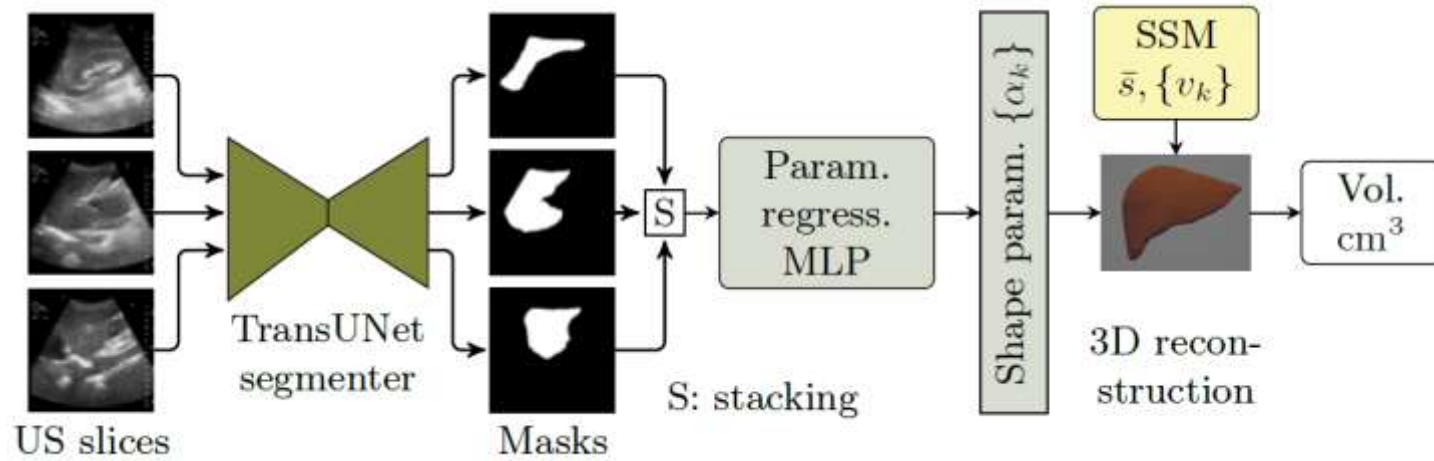


AR based Human Interactions for Automating a Robot Team



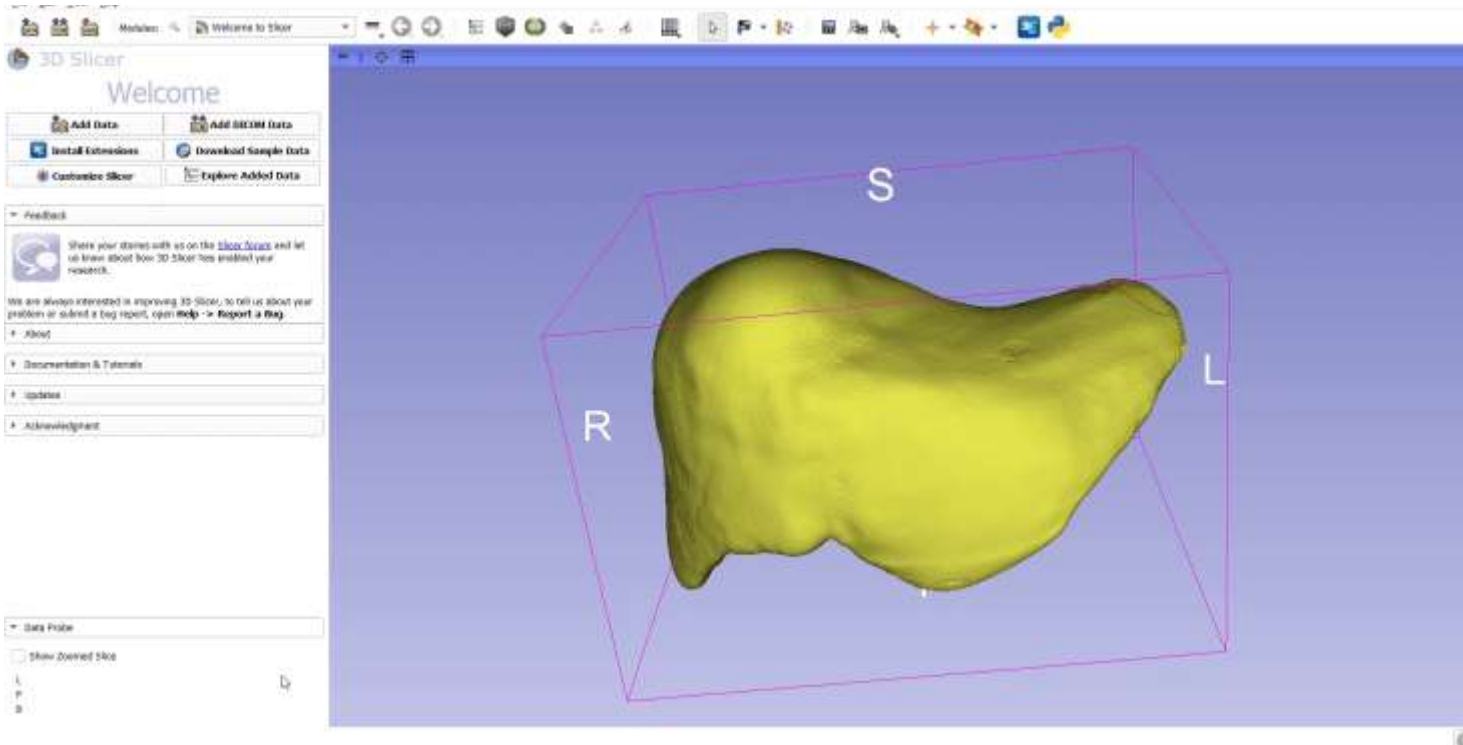
Dias, Adhitha, Hasitha Wellaboda, Yasod Rasanka, Menusha Munasinghe, Ranga Rodrigo, and Peshala Jayasekara. "Deep learning of augmented reality based human interactions for automating a robot team." In *2020 6th International Conference on Control, Automation and Robotics (ICCAR)*, pp. 175-182, 2020.

3D Reconstruction of the Liver with a Few US Scans



2 or 3 US scans
Volumetry possible

Kaushalay Sivayogaraj
Sahan Guruge
Jeevani Udupilille
Fariha Sitheeque
Saroj Jayasinghe
Gerard Fernando
Ranga Rodrigo
Rukshani Liyanaarachci



Patent Pending.

1. What is the primary difference between image processing and computer vision?
- A. Image processing enhances images; computer vision interprets them
 - B. Image processing is used only in photography; computer vision is used in robotics
 - C. Computer vision converts analog signals to digital; image processing doesn't
 - D. Image processing needs machine learning; computer vision doesn't
 - E. There is no difference

Which of the following is an example of a computer vision task?

- A. Histogram equalization
- B. Gaussian blur
- C. Semantic segmentation
- D. Image sharpening
- E. Padding

3. What is the goal of early vision in a computer vision pipeline?

- A. Compressing video
- B. Extracting high-level semantics from images
- C. Creating 3D point clouds
- D. Extracting low-level features like edges and textures
- E. Rendering synthetic scenes

4. Which of the following best describes the structure of this course (EN3160)?
- A. It focuses entirely on deep learning for image generation
 - B. It includes image processing, computer vision, and recent vision methods
 - C. It is limited to classical image processing
 - D. It covers only industrial inspection techniques
 - E. It is a hardware-oriented course
5. Which of the following is recommended for succeeding in the course?
- A. Memorize all the equations before exams
 - B. B. Implement at least one algorithm before the next class
 - C. C. Avoid reading papers; just follow lecture notes
 - D. D. Focus only on coding assignments
 - E. E. Use MATLAB and ignore Python
6. What is an advantage of using PyTorch in EN3160?
- A. It requires no coding knowledge
 - B. It is optimized only for image compression
 - C. It supports dynamic computation graphs and deep learning libraries
 - D. It is a hardware simulation tool
 - E. It only works with C++

7. Which of the following is a real-world application of computer vision?

- A. Applying a median filter to smooth an image
- B. Detecting faces in surveillance footage
- C. Printing high-resolution images
- D. Converting RGB to grayscale
- E. Saving images in compressed formats

8. What is an example of a generative model in vision?

- A. Histogram matching
- B. Optical flow estimation
- C. Diffusion models
- D. Sobel edge detection
- E. Bilinear interpolation

9. Which of the following is a typical use-case of segmentation in vision systems?

- A. Enhance the brightness of an image
- B. Estimate the motion of objects
- C. Detect boundaries between different regions in an image
- D. Convert images into JPEG format
- E. Add color to grayscale images

10. What is meant by "self-supervised pre-training" in deep learning for vision?
- A. Training a model on random images from the internet
 - B. Learning from labeled datasets with minimal supervision
 - C. Supervised training using multiple GPUs
 - D. Using reinforcement learning to train vision models
 - E. Learning meaningful representations from unlabeled data using surrogate tasks